



Aiden Stevenson Bradwell

Engineering Team Lead, AI Powered Full-Stack

Manchester, UK | aidenbradwell@gmail.com | +44 7354 829803

aidenbradwell.com | linkedin.com/in/aiden-bradwell | github.com/ABradwell

SUMMARY

An engineering lead with firm interests in project ownership, team empowerment, system modernization, and cloud architecture. I am a committed embracer of the modern development flow, and believe you can prioritise automation without losing great design or great people.

EXPERIENCE

Engineering Team Lead

Feb 2025 - Present

zally

Manchester, UK

- Architected **ML-Ops** support frameworks to productionise scalable **ONNX** pipelines
- Own **backup and disaster recovery**: **RTO/RPO** objectives encoded per platform area, coverage registers with drift detection, and recovery-point alarms
- Set the engineering standard through **17 operational runbooks** and **16 security design documents** used across the team
- Lead **architecture design decisions** on foundational system flows, translating non-functional requirements into actionable system designs
- Drove **AWS cost allocation** across the estate: service tagging for **EKS split cost**, per-stage cost categories, and automated teardown of review stacks
- Grew the team through regular one-to-ones, internal **skills matrices**, and review cycles targeted at individual growth areas
- Oversaw **demo applications and environments built to secure funding**, and engaged directly with investors on our core technologies and engineering methodology

Senior Software Engineer

Aug 2024 - Feb 2025

zally

Manchester, UK

- Built the **security CI/CD** gates: **BOLA** regression suites, **OWASP** OpenAPI conformance, **Trivy/OSV** CVE enforcement, and pre-commit secret scanning
- Designed and delivered an **organization-wide security suite** through CI/CD: CVE automation, **SBOM** generation, and injection protection across every repository
- Architected scalable streaming data pipelines on **Kafka** and **Apache Flink**, reaching across **AWS account boundaries**
- Increased pipeline throughput **50-fold**, holding performance through custom stress and load-testing harnesses feeding **Grafana** dashboards
- Migrated authentication from **Cognito** to **Keycloak** across portals, SDK token exchange, and pipeline authorizers, managing the stack for user federation and dashboard **RBAC**
- Implemented **Apache Iceberg** datalakes on **AWS Glue Schema** and **S3** storage
- Added small-file compaction and **data-tampering detection** through a collection of **Airflow** DAGs deployed within EKS

Software Engineer

zally

Jul 2023 - Aug 2024

Manchester, UK

- Founding engineer on the platform behind **Continuous Authentication**, and the largest contributor to its core repository
- Built the original **serverless stack** on **AWS** (Cognito, Lambda, Kinesis, DynamoDB) with **Node.js** services and **React Native** clients
- Designed and shipped modern **React** dashboards on **Node.js**, **shadcn/ui**, and the **SST** infrastructure-as-code framework
- Developed secure APIs documented in **OpenAPI 3.0**, with automated **drift detection** against the running services
- Built and maintained **PostgreSQL** databases on **AWS RDS** through the **Drizzle ORM**
- Developed the client-facing **SSE** connection system underpinning B2B integration flows
- Architected and implemented third-party **OAuth** token integrations

Java Web-App Developer (Spring)

Department of Fisheries and Oceans

May 2022 - Sep 2022

Ottawa, ON, Canada

- Developed a **REST API** using **Spring MVC** and **PostgreSQL** to store and manage ongoing prototypes and projects
- Developed a full-stack web application using **Spring Boot**, **PostgreSQL**, and the **Thymeleaf** template engine

Embedded Software Developer

Microchip Technology Incorporated

Jan 2021 - Aug 2021

Kanata, ON, Canada

- Reverse engineered legacy **automated testing software** in a semiconductor lab setting
- Developed a **Python CLI** with a **Robot Framework** automation back-end, **GPIO** instrument interaction, and oscilloscope output

Undergraduate Computer Vision Researcher

National Research Council of Canada

May 2020 - Aug 2020

Ottawa, ON, Canada

- Explored methods of early Alzheimer's detection through **computer vision** based eye-tracking
- Explored methods of RGB webcam **gaze estimation** and **eye tracking**

TECHNICAL SKILLS

Languages: TypeScript, Python, Java, Go, C/C++, SQL, Swift, Kotlin, Racket, Prolog, OCaml

Cloud & Platform: AWS, EKS/Kubernetes, Helm, ArgoCD, Istio, Terraform, SST, Airflow, Lambda, DynamoDB, RDS, S3, Kinesis, ECR, Docker

Data & Streaming: Apache Flink, Apache Kafka, Apache Iceberg, PostgreSQL, AWS Glue, Drizzle ORM, stream processing, throughput tuning

ML-Ops: TensorFlow, ONNX, FastAPI, AWS SageMaker, model serving, model versioning

Security: Keycloak, OIDC, OAuth 2.0, JWT, RBAC, Pod Security Standards, NetworkPolicy, BOLA testing, OWASP, Trivy, OSV, SBOM, gitleaks, WAF

Technologies: React, React Native, Node.js, Expo, shadcn/ui, Grafana, OpenTelemetry, OpenAPI 3.0, Spring Boot, computer vision

EDUCATION

Bachelor of Science, Major in Computer Science, Major in Psychology

University of Ottawa

Aug 2018 - Dec 2022

Ottawa, ON, Canada

- Graduated Magna Cum Laude (First-Class Honours equivalency)
- Member of the Dean's Honours List, 2018 to 2022